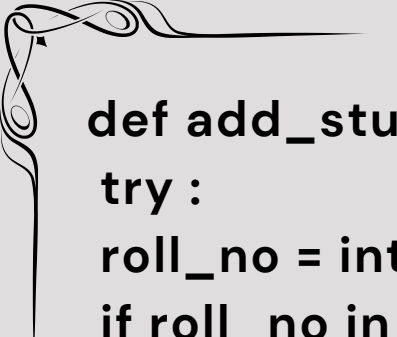


AI/ML PROJECT SUBMISSION



PROJECT OVERVIEW	STUDENT DETAILS
• Project Number: 1 • Task Name: Student Management System. • Difficulty: Easy	• Name: Pranav Gajanan Rane • College: Government Polytechnic Gondia • Branch: Computer Engineering • Domain: AI/ML



```
def add_student() : # Adds Students data
```

```
try :
```

```
roll_no = int(input("Enter the roll_no - "))
```

```
if roll_no in student_records :
```

```
print("Rollno already exists!")
```

```
else :
```

```
name = input("Enter your name - ")
```

```
if name.isdigit():
```

```
print("Name can only be in string!")
```

```
else :
```

```
mark1 = int(input("Enter your marks in Science - "))
```

```
mark2 = int(input("Enter your marks in Social Science  
- "))
```

```
mark3 = int(input("Enter your marks in Maths - "))
```

```
mark4 = int(input("Enter your marks in Hindi/Sanskrit  
- "))
```

```
mark5 = int(input("Enter your marks in English - "))
```

```
if (mark1 > 100 or mark2 > 100 or mark3 > 100 or  
mark4 > 100 or mark5 > 100) or \
```

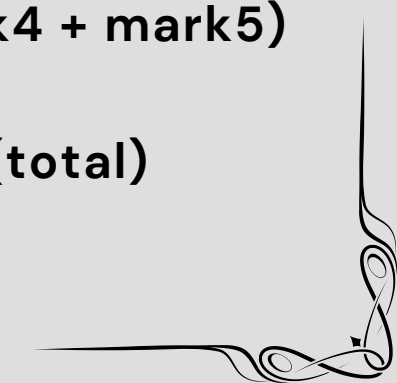
```
(mark1 < 0 or mark2 < 0 or mark3 < 0 or mark4 < 0  
or mark5 < 0):
```

```
print("Marks must be between 0 and 100!")
```

```
else:
```

```
total = (mark1 + mark2 + mark3 + mark4 + mark5)
```

```
percentage, grade = calculate_grade(total)
```



```
student_records[roll_no] = {  
    "name" : name,  
    "Science_Marks" : mark1,  
    "Social_Science_Marks" : mark2,  
    "Maths_Marks" : mark3,  
    "H/S_Marks" : mark4,  
    "English_Marks" : mark5,  
    "percentage" : percentage,  
    "grade" : grade  
}
```

```
except ValueError :  
    print("Enter correct Value!")
```

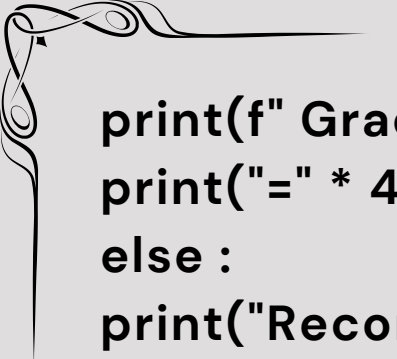
```
def view_all() : # Used to view all the students details  
if any.
```

```
    if not student_records :  
        print("Data doesn't exist")  
    else :  
        fmt = "{:<8} {:<15} {:<8} {:<8} {:<8} {:<8} {:<8} {:<12} {:  
<6}"  
        print("=" * 105)  
        print(fmt.format("Roll No", "Name", "Sci", "SocSci",  
"Math", "Hin/San", "Eng", "Percentage", "Grade"))  
        print("=" * 105)
```

```
for roll_no,details in student_records.items() :  
    print(fmt.format(  
        roll_no,  
        details["name"],  
        details["Science_Marks"],  
        details["Social_Science_Marks"],
```

```
details["Maths_Marks"],
details["H/S_Marks"],
details["English_Marks"],
f"{details['percentage']:.2f}%", # rounds percentage
upto 2 decimal places
details["grade"]
))
print("=" * 105)
```

```
def search_student() : # Searches student a student
by their roll_no
try :
if not student_records :
print("Data doesn't exist")
else :
roll_no = int(input("Enter a rollno to view its data - "))
if roll_no in student_records :
details = student_records[roll_no]
print("\n" + "=" * 40)
print(f" FOUND RECORD (Roll No: {roll_no})")
    print("=" * 40)
    print(f" Name:  {details['name']}")
    print(f" Science: {details['Science_Marks']}")
                    print(f"      Social      Sci:
{details['Social_Science_Marks']}")
    print(f" Maths:  {details['Maths_Marks']}")
    print(f" Hindi/Sans: {details['H/S_Marks']}")
    print(f" English: {details['English_Marks']}")
    print("-" * 40)
    print(f" Percentage: {details['percentage']:.2f}%")
```



```
print(f" Grade: {details['grade']}")
print("=" * 40)
else :
print("Record not found for this key!")
except :
print("Pls enter correct values!")
```

```
def delete_student() : # Removes a students data
    try :
        roll_no = int(input("Enter the roll no whose data you
want to remove - "))
        if roll_no in student_records :
            print(f"Removing roll_no {roll_no}'s data.")
            del student_records[roll_no]
            print("Data Removed!")
        else :
            print("Roll no does not exists")
    except ValueError :
        print("Pls enter correct value!")
```

```
def calculate_grade(total) : # calculates grade and
send back to add add_student()
    percentage = (total / 500) * 100
    grade = ""
    if percentage >= 90 :
        grade = "A"
    elif percentage >= 80 :
        grade = "B"
    elif percentage >= 70 :
```



```
grade = "C"
elif percentage >= 60 :
grade = "D"
elif percentage >= 50 :
grade = "E"
else :
grade = "F"
return percentage,grade
def update_student() : #updates any required values
of student
try :
roll_no = int(input("Enter a rollno to change its data
- "))
if roll_no in student_records :

print("What do you want to change - ")
print("1)Name")
print("2)Marks of Science")
print("3)Marks of Math")
print("4)Marks of Social Science")
print("5)Marks of Hindi/Sanskrit")
print("6)Marks of English ")

choice = input("Enter your choice (1 - 6) -")
if choice == "1" :
name = input("Change the name to - ")
if name.isdigit():
print("Name can only be in string!")
else :
student_records[roll_no]['name'] = name
print("Name updated")
```



```
def update_student() : #updates any required values  
of student
```

```
try :
```

```
    roll_no = int(input("Enter a rollno to change its data  
- "))
```

```
    if roll_no in student_records :
```

```
        print("What do you want to change - ")
```

```
        print("1)Name")
```

```
        print("2)Marks of Science")
```

```
        print("3)Marks of Math")
```

```
        print("4)Marks of Social Science")
```

```
        print("5)Marks of Hindi/Sanskrit")
```

```
        print("6)Marks of English ")
```

```
    choice = input("Enter your choice (1 - 6) -")
```

```
    if choice == "1" :
```

```
        name = input("Change the name to - ")
```

```
        if name.isdigit():
```

```
            print("Name can only be in string!")
```

```
        else :
```

```
            student_records[roll_no]['name'] = name
```

```
            print("Name updated")
```

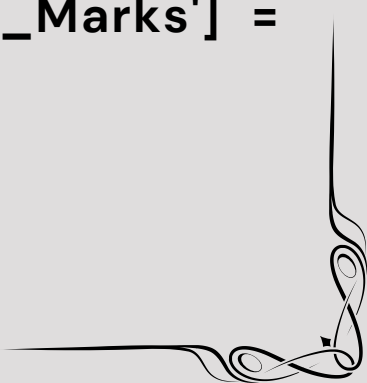
```
    elif choice == "2" :
```

```
        mark1 = int(input("Enter updated marks - "))
```

```
            student_records[roll_no]['Science_Marks'] =
```

```
mark1
```

```
            print("Marks Updated!")
```



elif choice == "3" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['Maths_Marks'] = mark1

print("Marks Updated!")

elif choice == "4" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]

['Social_Science_Marks'] = mark1

print("Marks Updated!")

elif choice == "5" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['H/S_Marks'] = mark1

print("Marks Updated!")

elif choice == "6" :

mark1 = int(input("Enter updated marks - "))

student_records[roll_no]['English_Marks'] =

mark1

print("Marks Updated!")

else :

print("Pls Enter the correct value(1-6)")

total = student_records[roll_no]['Science_Marks']

+ student_records[roll_no]['Social_Science_Marks'] +

student_records[roll_no]['Maths_Marks'] +

student_records[roll_no]['English_Marks'] +

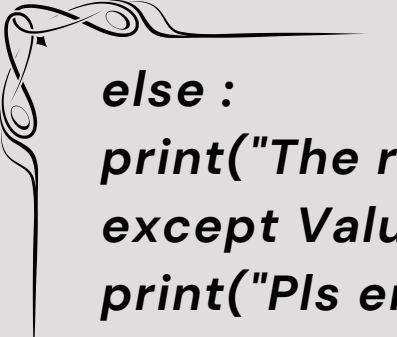
student_records[roll_no]['H/S_Marks']

percentage, grade = calculate_grade(total)

student_records[roll_no]['percentage'] =

percentage

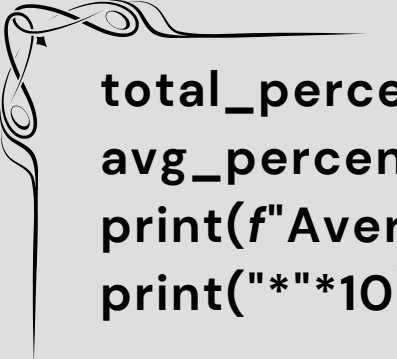
student_records[roll_no]['grade'] = grade



```
else :  
print("The rollno doesnt exist!")  
except ValueError :  
print("Pls enter correct values!")
```

```
def class_report(): # Bonus Task  
    if not student_records:  
        print("No data to generate report!")  
        return  
    else:  
        #Calculating topper  
        highest = -1.0  
        name = ""  
  
        for key,value in student_records.items() :  
            pct = value['percentage']  
  
            if pct > highest :  
                highest = pct  
                name = value['name']  
        print("*"*10)  
        print(f"{name} achieved highest percentage -  
{highest}")  
        print("*"*10)  
  
        #Finding Average Percentage  
        all_students = len(student_records)  
        total_percentage = 0  
        for key,value in student_records.items():
```





```
total_percentage += value['percentage']
avg_percentage = total_percentage / all_students
print(f"Average Percentage are - {avg_percentage}")
print("*"*10)
```

```
#Pass/Fail
```

```
passed = 0
```

```
failed = 0
```

```
for key,value in student_records.items():
```

```
    if value['percentage'] < 50 :
```

```
        failed += 1
```

```
    else :
```

```
        passed += 1
```

```
print(f"Total student passed are - {passed}")
```

```
print(f"Total Student Failed are - {failed}")
```

```
def show_menu() : #Shows what you can perform
```

```
    print("What you can perfrom - ")
```

```
    print("1)Add a Student")
```

```
    print("2)View All Students")
```

```
    print("3)Search a student by their Roll Number")
```

```
    print("4)Update a student record")
```

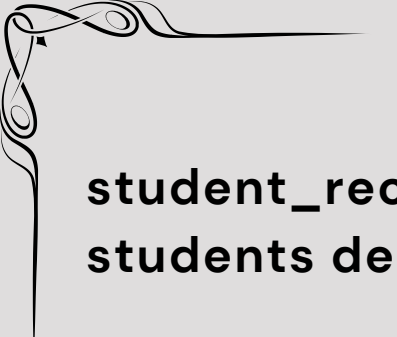
```
    print("5)Remove a student data")
```

```
    print("6)See Class Report")
```

```
    print("7)Exit")
```

```
'''Actual System Below'''
```





```
student_records = {} #(Global Declaration) Holds  
students details
```

```
def main():
```

```
    while True:
```

```
        try:
```

```
            print("-" * 58)
```

```
            print("_____-Welcome to Student Management  
System!_____")
```

```
            print("Note - Asks for user input to perform a  
specific action.")
```

```
            print("-" * 58)
```

```
        show_menu()
```

```
        choice = input("Enter what you want to do (1 - 7) -  
")
```

```
        print("-" * 30)
```

```
        if choice == "1":
```

```
            add_student()
```

```
        elif choice == "2":
```

```
            view_all()
```

```
        elif choice == "3":
```

```
            search_student()
```

```
        elif choice == "4":
```

```
            update_student()
```

```
        elif choice == "5":
```

```
            delete_student()
```

```
        elif choice == "6":
```

```
class_report()
elif choice == "7" :
print("Exited Successfully!")
break
else :
print("Pls enter a valid choice(1 - 7)")
except ValueError :
print("Enter correct value!")

if __name__ == "__main__":
main()
```

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached

TEST OUTPUTS -

```
-----
_____Welcome to Student Management System!_____
Note - Asks for user input to perform a specific action.
-----

What you can perfrom -
1)Add a Student
2)View All Students
3)Search a student by their Roll Number
4)Update a student record
5)Remove a student data
6)See Class Report
7)Exit
Enter what you want to do (1 - 7) - █
```

Enter what you want to do (1 - 7) - 1

Enter the roll_no - 101
Enter your name - Pranav
Enter your marks in Science - 98
Enter your marks in Social Science - 97
Enter your marks in Maths - 96
Enter your marks in Hindi/Sanskrit - 95
Enter your marks in English - 85

Enter what you want to do (1 - 7) - 2

Roll No	Name	Sci	SocSci	Math	Hin/San	Eng	Percentage	Grade
101	Pranav	98	97	96	95	85	94.20%	A
102	Gauri	95	75	85	95	75	85.00%	B
103	Ishant	23	42	32	16	23	27.20%	F

Enter what you want to do (1 - 7) - 3

Enter a rollno to view its data - 101

=====

FOUND RECORD (Roll No: 101)

=====

Name: Pranav
Science: 98
Social Sci: 97
Maths: 96
Hindi/Sans: 95
English: 85

Percentage: 94.20%
Grade: A

=====



```
Enter what you want to do (1 - 7) - 6
-----
*****
Pranav achieved highest percentage - 94.19999999999999
*****
Average Percentage are - 68.8
*****
Total student passed are - 2
Total Student Failed are - 1
```

TEST OUTPUTS IN BROKEN CASES -

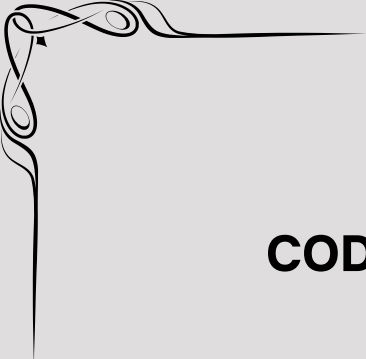
```
Enter what you want to do (1 - 7) - 1
-----
Enter the roll_no - Pranav
Enter correct Value!
```

```
Enter what you want to do (1 - 7) - 3
-----
Enter a rollno to view its data - 95
Record not found for this key!
```

```
Enter what you want to do (1 - 7) - 5
-----
Enter the roll no whose data you want to remove - 95
Roll no does not exists
```

```
Enter what you want to do (1 - 7) - 5
-----
Enter the roll no whose data you want to remove - Pranav
Pls enter correct value!
-----
```





CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED

